Balancing Equations #1

1. C_3H_8 + O_2 -------> CO_2 + H_2O

2. $Al_2(SO_3)_3$ + $NaOH$ ------> Na_2SO_3 + $Al(OH)_3$

3. Al_2O_3 + Fe ------> Fe_3O_4 + Al

4. $KClO_3$ ------> KCl + O_2

5. NH_4NO_3 -------> N_2O + H_2O

6. $NaHCO_3$ -------> Na_2CO_3 + H_2O + CO_2

7. P_4O_{10} + H_2O ------> H_3PO_4

8. Al + H_2SO_4 -------> $Al_2(SO_4)_3$ + H_2

9. Be_2C + H_2O -------> $Be(OH)_2$ + CH_4

10.*** S + HNO_3 -------> H_2SO_4 + NO_2 + H_2O

11. NH_3 + CuO -------> Cu + N_2 + H_2O

12.*** Cu + HNO_3 -------> $Cu(NO_3)_2$ + NO + H_2O

Student Worksheet 5.2A

Extra Practice Questions: Balancing Chemical Equations

Balance the following equations by inspection.

1. $\_Ni_{(s)} + \_HCl_{(aq)} \rightarrow \_NiCl_{2(aq)} + \_H_{2(g)}$

2. $\_Ca(OH)_{2(s)} + \_HCl_{(aq)} \rightarrow \_CaCl_{2(aq)} + \_HOH_{(l)}$

3. $\_Cl_{2(g)} + \_NaBr_{(aq)} \rightarrow \_Br_{2(l)} + \_NaCl_{(aq)}$

4. $\_Cr_2O_{3(s)} \rightarrow \_Cr_{(s)} + \_O_{2(g)}$

5. $\_Fe_{(s)} + \_HCl_{(aq)} \rightarrow \_FeCl_{3(aq)} + \_H_{2(g)}$

6. $\_C_3H_{6(g)} + \_O_{2(g)} \rightarrow \_CO_{2(g)} + \_H_2O_{(g)}$

7. $\_P_{4(s)} + \_F_{2(g)} \rightarrow \_PF_{3(l)}$

8. $\_Ca(NO_3)_{2(aq)} + \_KOH_{(aq)} \rightarrow \_Ca(OH)_{2(s)} + \_KNO_{3(aq)}$

9. $\_KHCO_{3(s)} \rightarrow \_K_2CO_{3(s)} + \_H_2O_{(l)} + \_CO_{2(g)}$

10. $\_H_3PO_{4(aq)} + \_NaOH_{(aq)} \rightarrow \_Na_3PO_{4(aq)} + \_HOH_{(l)}$

11. $\_Ca(NO_3)_{2(aq)} + \_Na_3PO_{4(aq)} \rightarrow \_Ca_3(PO_4)_{2(s)} + \_NaNO_{3(aq)}$

12. $\_Cu_{(s)} + \_HNO_{3(aq)} \rightarrow \_Cu(NO_3)_{2(aq)} + \_NO_{2(g)} + \_H_2O_{(l)}$

Balancing Equations Race

1) ___ C_3H_8 + ___ O_2 → ___ CO_2 + ___ H_2O

2) ___ Al + ___ Fe_3N_2 → ___ AlN + ___ Fe

3) ___ Na + ___ Cl_2 → ___ NaCl

4) ___ H_2O_2 → ___ H_2O + ___ O_2

5) ___ $C_6H_{12}O_6$ + ___ O_2 → ___ H_2O + ___ CO_2

6) ___ H_2O + ___ CO_2 → ___ C_7H_8 + ___ O_2

7) ___ $NaClO_3$ → ___ NaCl + ___ O_2

8) ___ $(NH_4)_3PO_4$ + ___ $Pb(NO_3)_4$ → ___ $Pb_3(PO_4)_4$ + ___ NH_4NO_3

9) ___ BF_3 + ___ Li_2SO_3 → ___ $B_2(SO_3)_3$ + ___ LiF

10) ___ C_7H_{17} + ___ O_2 → ___ CO_2 + ___ H_2O

11) ___ $CaCO_3$ + ___ H_3PO_4 → ___ $Ca_3(PO_4)_2$ + ___ H_2CO_3

12) ___ Ag_2S → ___ Ag + ___ S_8

13) ___ KBr + ___ $Fe(OH)_3$ → ___ KOH + ___ $FeBr_3$

14) ___ KNO_3 + ___ H_2CO_3 → ___ K_2CO_3 + ___ HNO_3

15) ___ $Pb(OH)_4$ + ___ Cu_2O → ___ PbO_2 + ___ CuOH

16) ___ $Cr(NO_2)_2$ + ___ $(NH_4)_2SO_4$ → ___ $CrSO_4$ + ___ NH_4NO_2

17) ___ KOH + ___ $Co_3(PO_4)_2$ → ___ K_3PO_4 + ___ $Co(OH)_2$

18) ___ $Sn(NO_2)_4$ + ___ Pt_3N_4 → ___ Sn_3N_4 + ___ $Pt(NO_2)_4$

19) ___ B_2Br_6 + ___ HNO_3 → ___ $B(NO_3)_3$ + ___ HBr

20) ___ ZnS + ___ AlP → ___ Zn_3P_2 + ___ Al_2S_3

WRITE A BALANCED EQUATION FOR EACH OF THE FOLLOWING:

1. Calcium carbonate yields Calcium oxide plus carbon dioxide (CO_2)

2. Aluminum plus cupric nitrate yields aluminum nitrate plus copper.

3. Chlorine plus potassium iodide yields iodine plus potassium chloride.

4. Silver nitrate plus barium chloride yields barium nitrate plus silver chloride.

5. Mercuric oxide yields mercury plus oxygen.

6. Potassium plus water yields hydrogen plus potassium hydroxide.

7. Zinc plus sulfur (S_8) yields zinc sulfide.

8. Iron plus oxygen yields ferric oxide.

9. Chromium plus chlorine yields chromic chloride.

10. Silver plus sodium chromate yields silver chromate plus sodium.

SCH-3A Practice Writing Chemical Equations

Translate the following descriptions into a chemical equation using appropriate notations. Balance these and determine what type of reaction each one is.

1. A solid piece of iron oxidizes to form rust. Iron(II) oxide.
2. A solution of nitric acid dissolves solid copper, forming a solution of copper (II) nitrate and hydrogen gas.
3. Solutions of sulphuric acid and sodium hydroxide neutralize one another forming a solution of sodium sulphate and water.
4. When zinc is dropped into hydrochloric acid, hydrogen gas forms as well as a solution of zinc chloride.
5. When solutions of lead (II) nitrate and potassium iodide are mixed, a predicate of lead (II) iodide and a solution of potassium nitrate forms.
6. When magnesium is heated in a strong flame, it oxidizes to magnesium oxide.
7. When an electric current is passed through molten sodium chloride, sodium metal and chlorine gas forms.
8. When Zinc and sulphur are mixed and heated in a Bunsen flame, solid zinc sulphide forms.
9. When copper(II) sulphate pentahydrate is heated, anhydrous copper (II) sulphate and water form.
10. When solutions of silver nitrate and hydrochloric acid are mixed, a white solid, silver chloride and nitric acid form.
11. When hydrogen gas and chlorine gas are mixed and treated with ultraviolet light, hydrogen chloride gas forms explosively.
12. When phosphorus is exposed to oxygen, it ignites forming diphosphorus pentaoxide.
13. When sodium metal is added to water, sodium hydroxide and hydrogen gas form.
14. When sodium carbonate is added to hydrochloric acid, carbon dioxide gas forms as well as water and a solution of sodium chloride.
15. When hydrogen and oxygen are ignited with a spark, water forms explosively.